Microbiology: Urinary Tract Infections

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Session Objectives

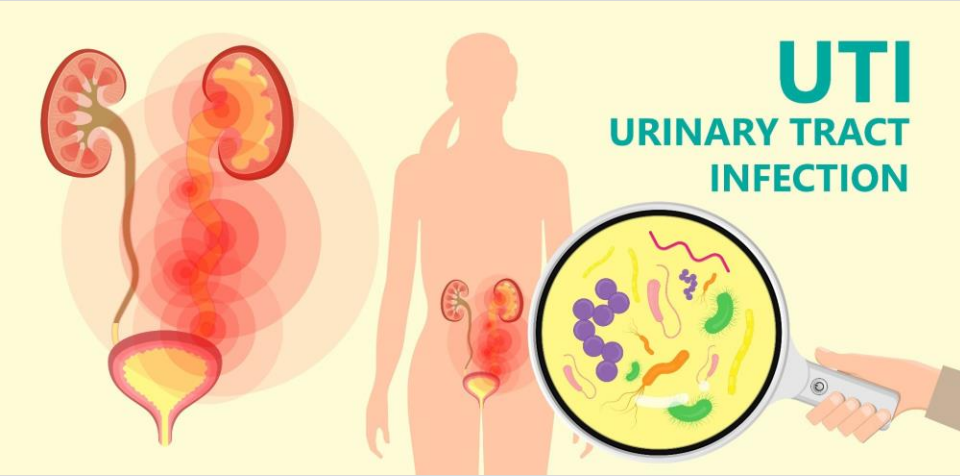
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1. Describe the epidemiology, risk factors, and terminology associated with urinary tract infections (UTIs).
2. Explain UTI pathogenesis and how it leads to different types of diseases.
3. Describe common symptoms associated with UTIs and diagnostic approaches used for identification.
4. Differentiate the pathogens that cause urinary tract infections.

Introduction

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- UTIs are the most frequent outpatient infection*
- Pathogenesis
  - Colonization and ascension via the urethra
  - Subsequent ascension via the ureter
- More common in women
- Risk factors:
  - Sexual activity, age, previous UTIs, medical devices, etc.

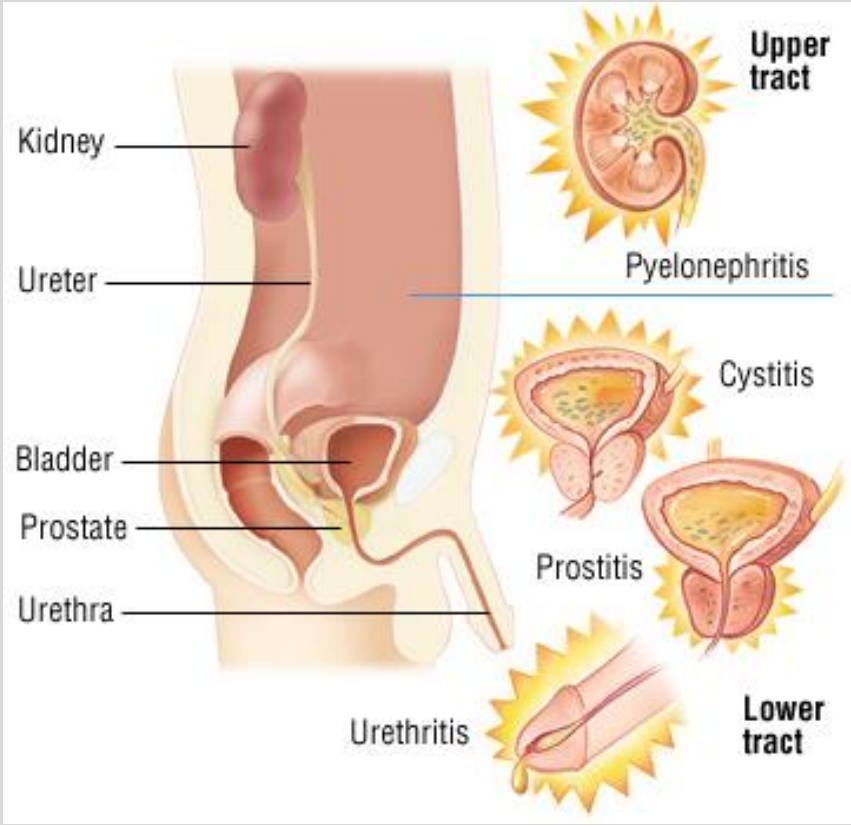


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*Medina M, Castillo-Pino E. An introduction to the epidemiology and burden of urinary tract infections. Ther Adv Urol. 2019 May 2;11:1756287219832172. doi: 10.1177/1756287219832172. PMID: 31105774; PMCID: PMC6502976.

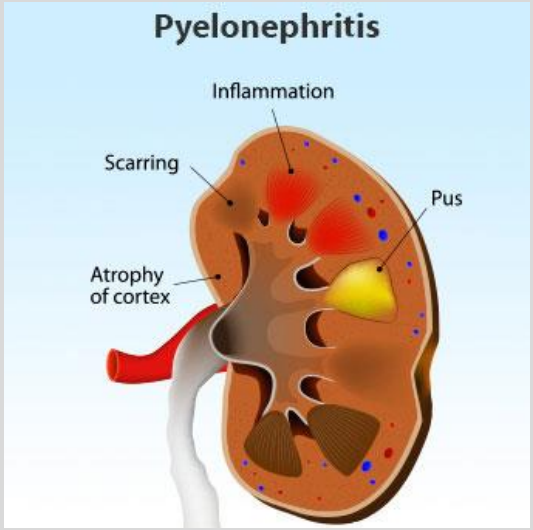
Types of UTIs

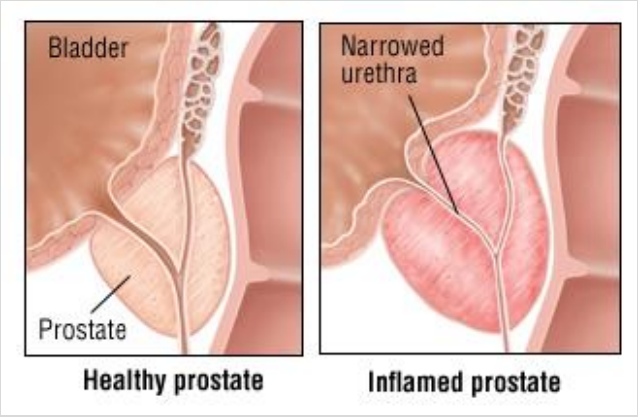
- Defined by anatomical structure:
  - Urethritis, Cystitis, Pyelonephritis, Prostatitis
- Asymptomatic vs. Symptomatic
  - *(Asymptomatic bacteriuria: asymptomatic with 2 consecutive (+) urine cultures with the same organism)*
- Complicated vs. uncomplicated
  - Uncomplicated UTI: infections in healthy women
  - Complicated UTI: all other types of UTI



UTI Presentation

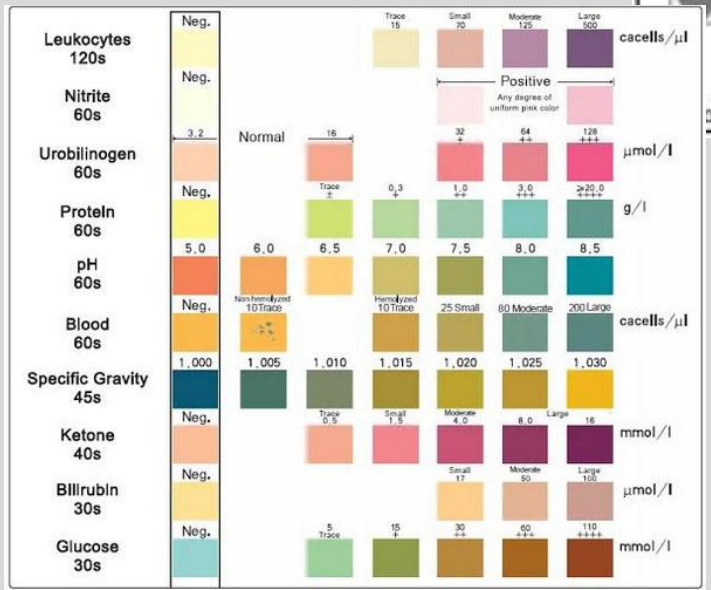
- Urethritis:
  - Gonococcal vs. Non-gonococcal (NGU)
  - Dysuria, urethral discharge, or both
- Cystitis:
  - Inflammation of the mucosal surface bladder
  - Dysuria, frequency, urgency, suprapubic pain
- Pyelonephritis:
  - Infection/inflammation of the kidney
  - Flank pain, fever, costovertebral tenderness
- Prostatitis:
  - Inflammation of the prostate
  - Acute vs chronic

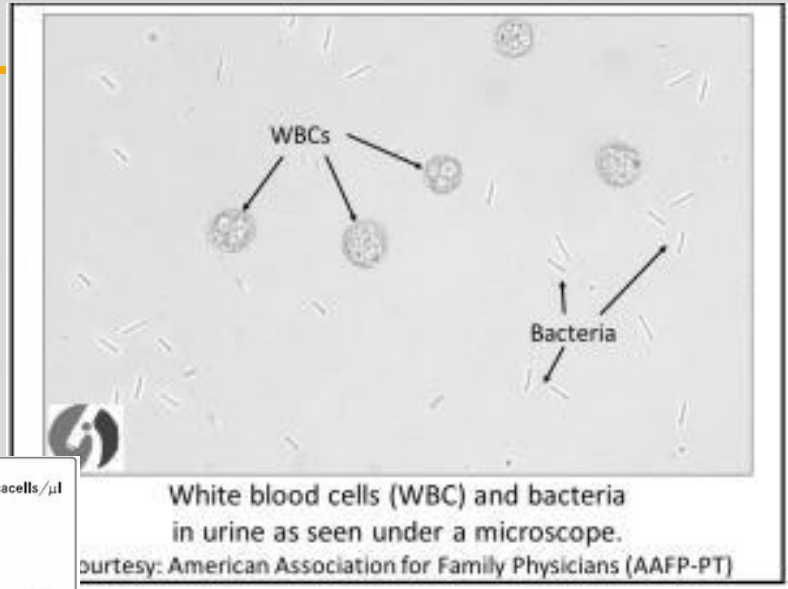




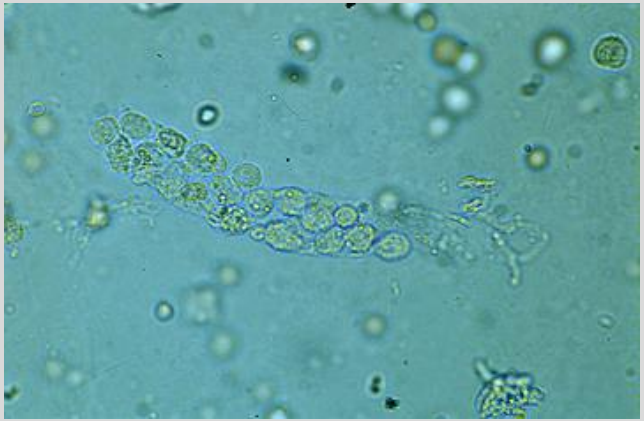
Diagnosis

- Clinical presentation
- Lab tests:
  - Dipstick
    - Leukocyte esterase
    - Nitrite
  - Microscopy
    - Pyuria
    - Hematuria
  - Culture





White blood cells (WBC) and bacteria in urine as seen under a microscope.
Courtesy: American Association for Family Physicians (AAFP-PT)



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Diagnosis

- Blood agar
  - Hemolysis



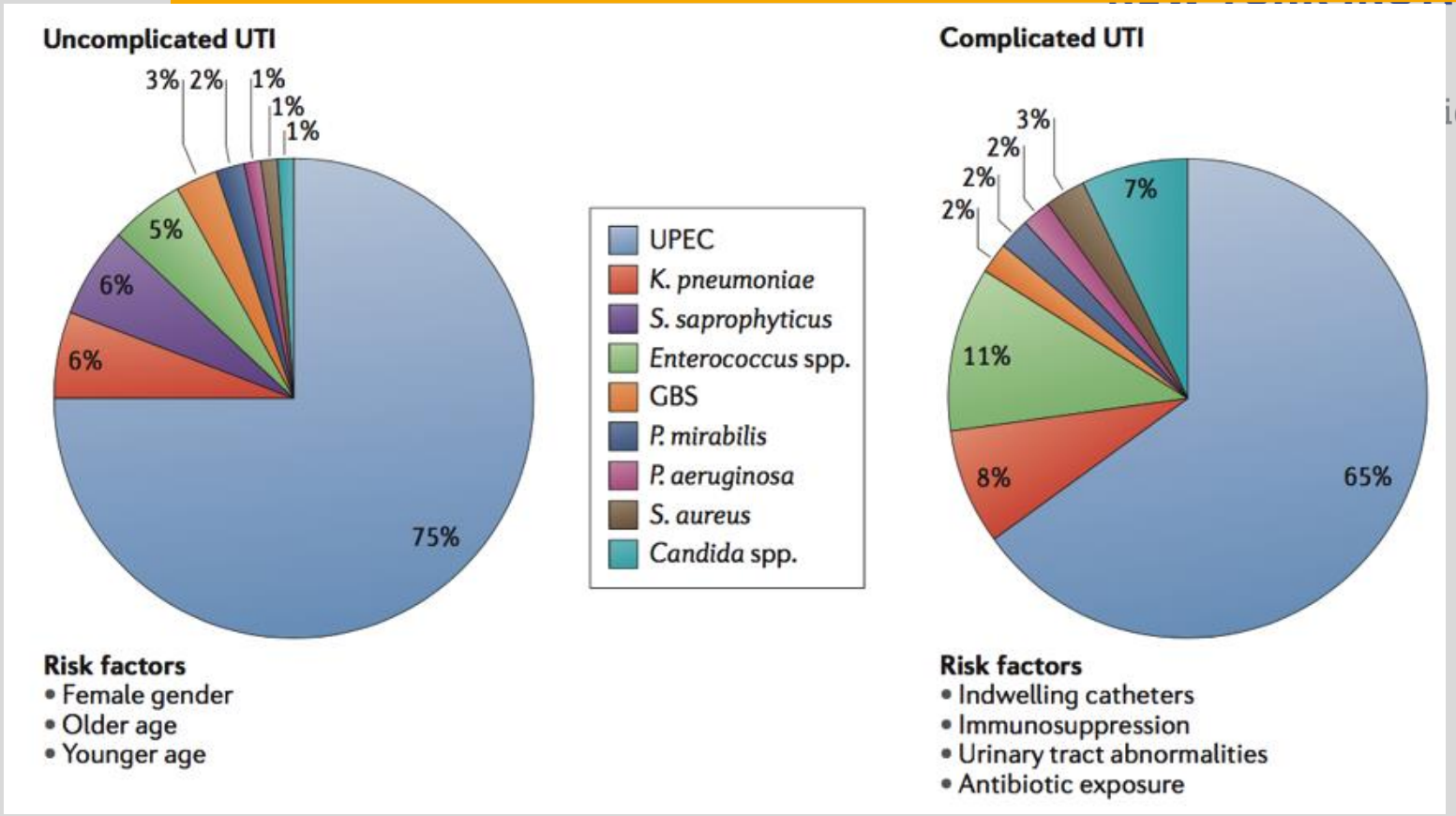
- MacConkey agar
  - Enteric Gram negative
  - Lactose fermentation



- Chocolate agar
  - Fastidious species



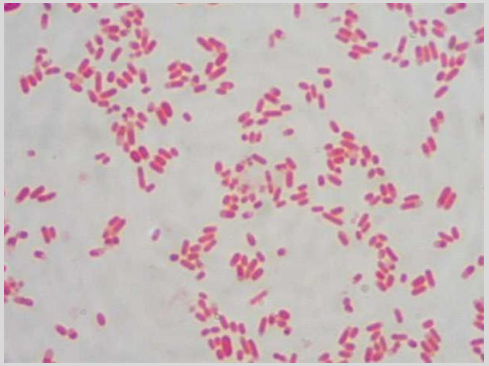
Etiology of UTIs

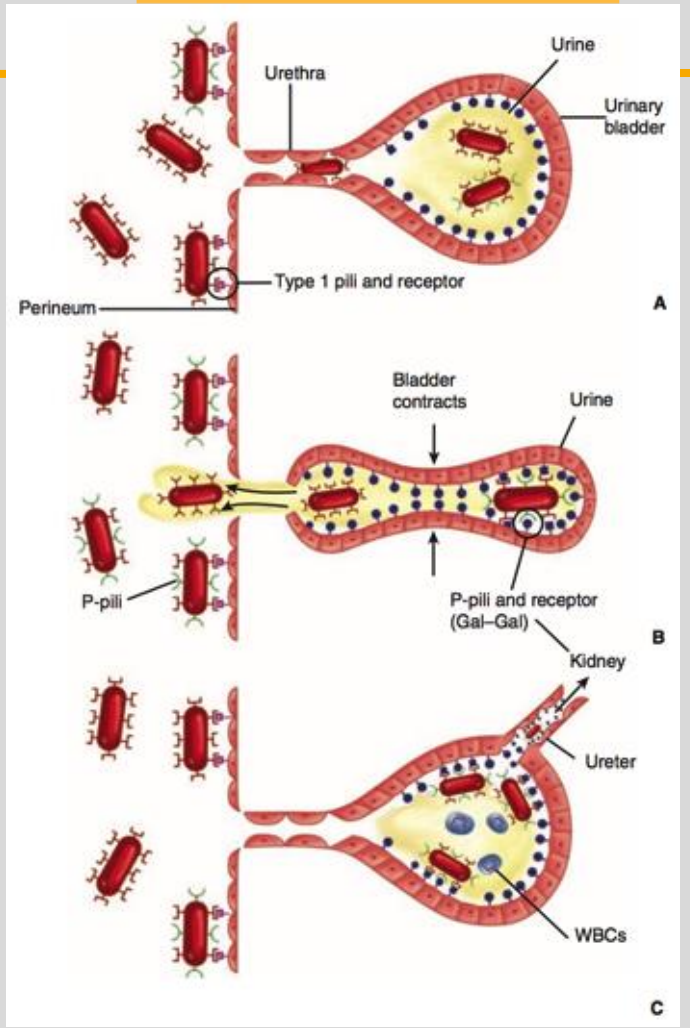


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Escherichia coli

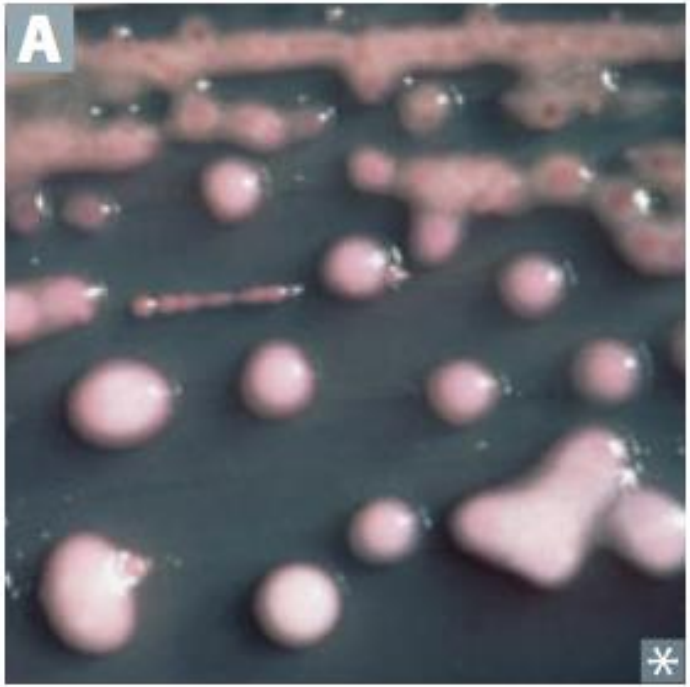
- Most common cause of UTIs
- Gram negative enteric bacilli
- Distinguishing features:
  - Lactose fermenter
  - IMViC test (++--)
  - Motile
- Uropathogenic *E. coli* (UPEC)
  - Virulence factors:
    - "P"-pili and Type 1 fimbriae
    - Toxins
    - Iron acquisition
    - Capsule





Klebsiella pneumoniae

- Gram negative enteric bacilli
- Distinguishing features:
  - Mucoid and viscous colonies
  - Lactose-fermenter
  - Encapsulated
  - Non-motile



Proteus mirabilis

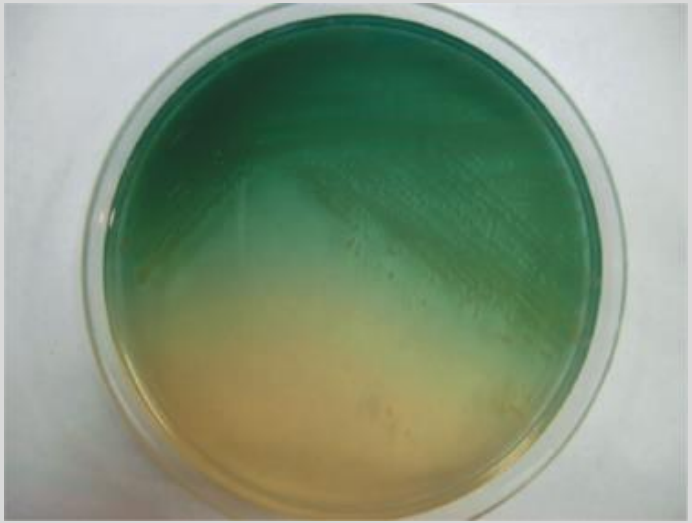


Image: http://atlas.sund.ku.dk/microatlas/veterinary/bacteria/Proteus_mirabilis/

- Gram negative enteric bacilli
- Distinguishing features:
  - Urease +; leads to struvite and apatite stones
  - Highly Motile: “Swarming” appearance on media
  - Does not ferment lactose

Pseudomonas aeruginosa

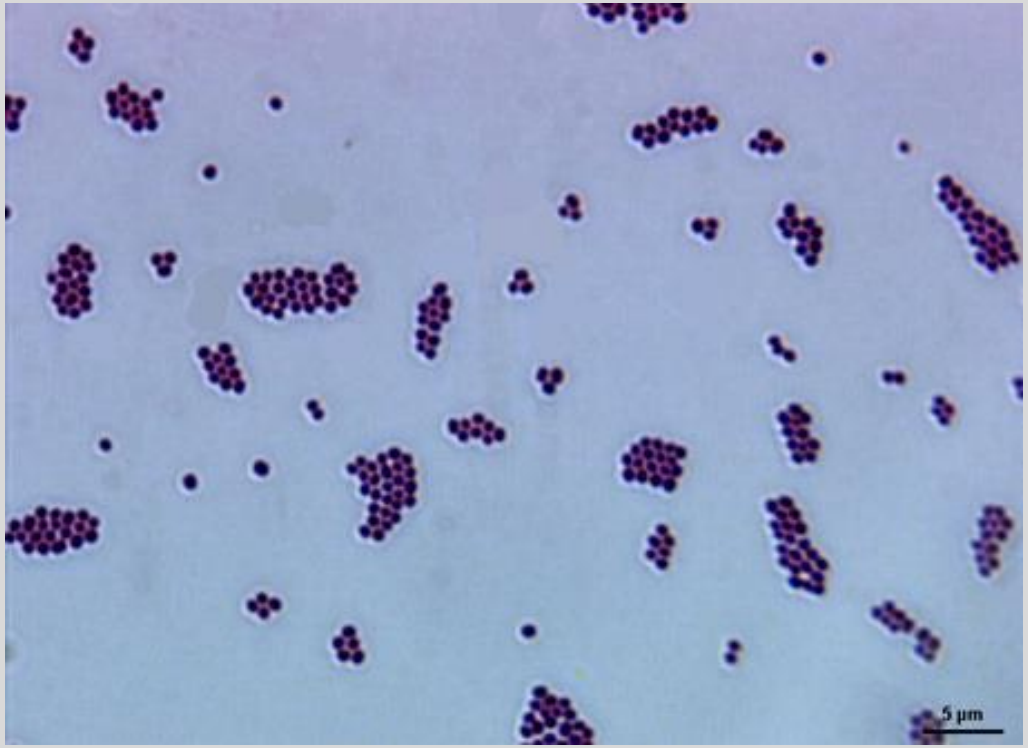
- Gram negative bacilli
- Distinguishing features:
  - Aerobic
  - Pyocyanin (blue-green pigment) →
  - Oxidase +
  - Encapsulated
  - Motile
  - Non lactose fermenter
  - Grape-like odor



Staphylococcus saprophyticus

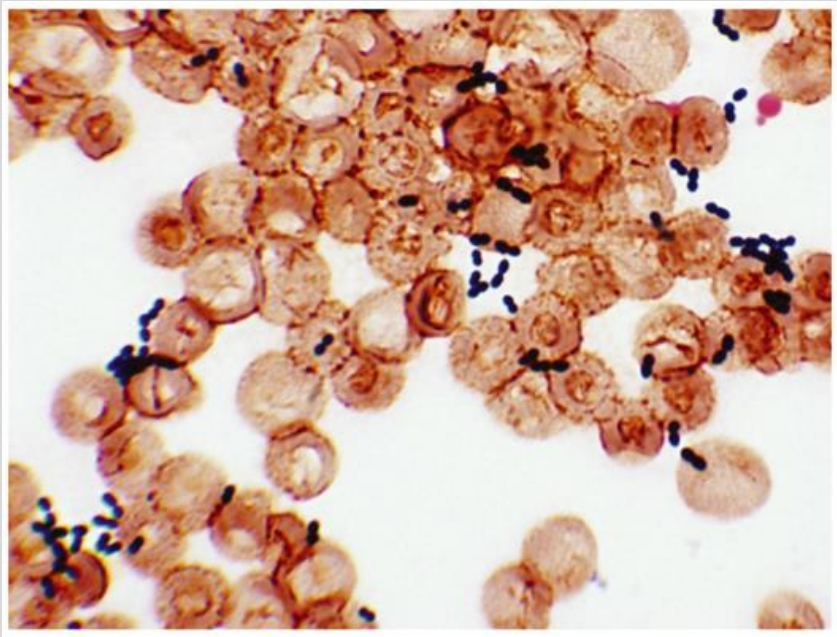
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- Gram positive coccus in clusters
- Distinguishing features:
  - Catalase positive
  - Coagulase negative
  - Gamma-hemolytic
  - Novobiocin resistant



Enterococcus spp.

- *E. faecalis & E. faecium*
- Gram positive enteric cocci
- Often drug resistant and nosocomial
- Distinguishing features:
  - Growth in 6.5% NaCl
  - Bile Esculin +
  - Catalase negative
  - Mostly non-hemolytic or α-hemolytic
  - Optochin resistant



Candida spp.

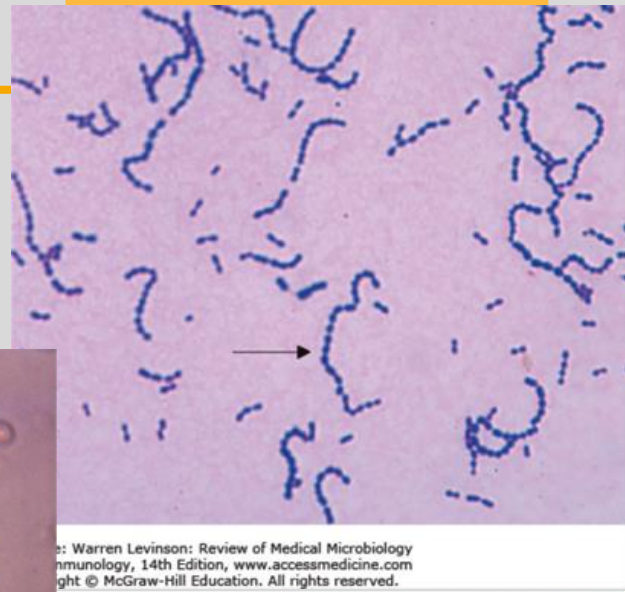
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- Opportunistic fungus
  - *C. albicans*
- Hospital associated infections
- Risk factors:
  - Medical devices
  - Antibiotic use
  - Diabetes
- Distinguishing features:
  - Pseudohyphae
  - Germ tube

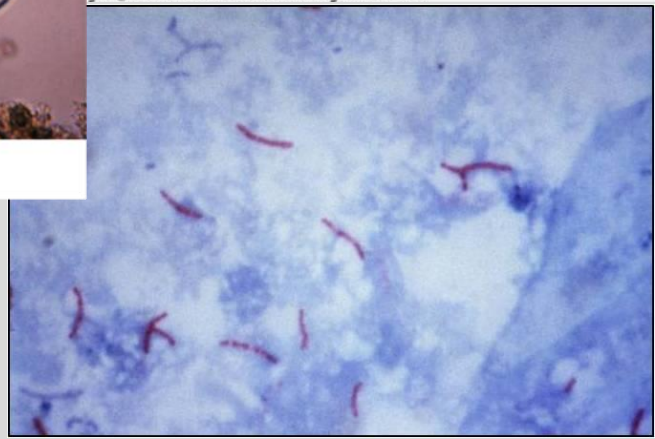


Other Possible Causes of UTIs

- *Streptococcus* spp.
  - *GBS, GAS*
- *Staphylococcus aureus*
- *Mycobacterium tuberculosis*
- Enterobacteriaceae
- *Corynebacterium urealyticum*
- *Schistosoma haematobium*
- Viruses
  - Polyomavirus, adenovirus

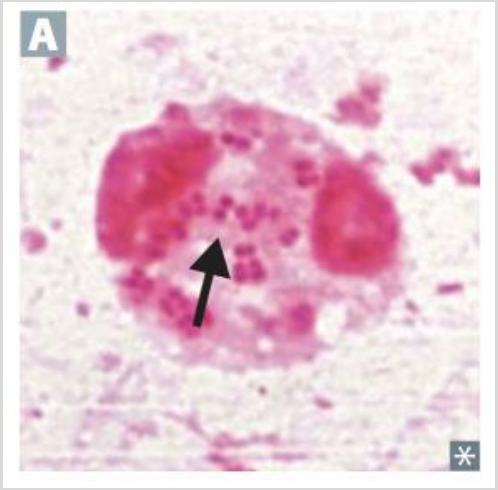


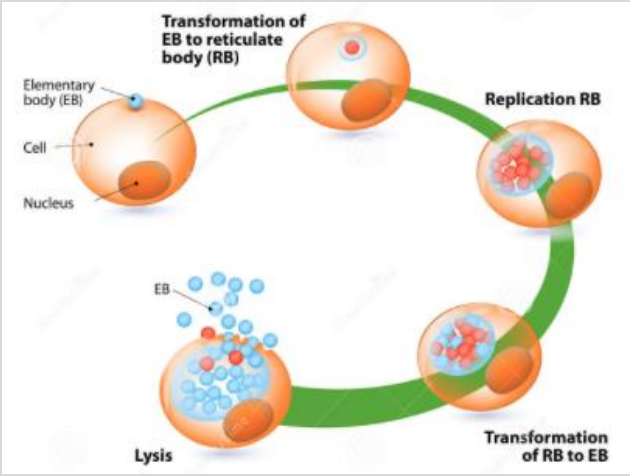


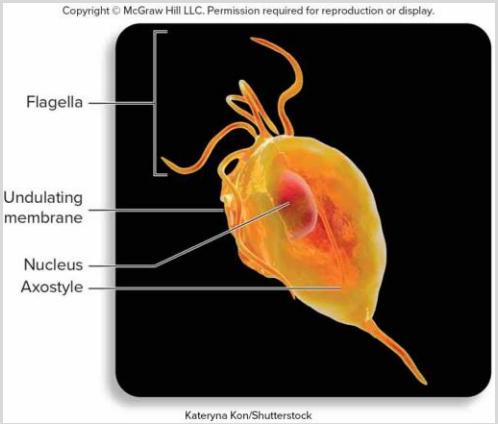


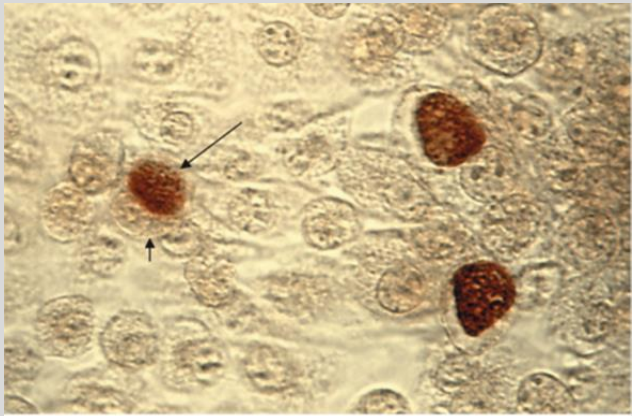
Possible Causes of Urethritis

- *Neisseria gonorrhoeae*
- *Chlamydia trachomatis*
- *Mycoplasma genitalium*
- *Ureaplasma urealyticum*
- *Trichomonas vaginalis*









Source: Warren Levinson: Review of Medical Microbiology and Immunology, 14th Edition, www.accessmedicine.com
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Summary

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- Urinary tract infections (UTIs) will present in different ways based on the type and anatomical location
- Uropathogens are usually part of the fecal flora that colonize and ascend the genitourinary tract
- The most common cause of UTIs is uropathogenic *E. coli* (UPEC)
- A small number of microbes with distinct characteristics cause the vast majority of UTIs.
- A minority of UTIs can be caused by a wide variety of microbes, including bacteria, viruses, and parasites.

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